

HOSPITAL MANAGEMENT SYSTEM (HMS)

PROJECT PROPOSAL DOCUMENT – PREMIUM VERSION

1. PROPOSAL SUMMARY

We propose to develop a **Premium Hospital Management System (HMS)** — a complete digital healthcare ecosystem with enterprise web platform, mobile applications (Patient + Doctor), AI-powered features, full EMR/EHR, telemedicine, and enterprise-level automation. Designed for **large hospitals and multi-branch healthcare organizations (30+ doctors)** requiring high scalability, automation, security, and performance.

Ideal For: Large hospitals, multi-branch healthcare chains, super-specialty centers (30+ doctors)

2. OBJECTIVES

- Fully automate all hospital operations end-to-end across departments
- Provide seamless digital experience for patients, doctors, and staff
- Enable real-time multi-branch data access and analytics
- Implement AI-driven smart systems for diagnostics, attendance, and insights
- Ensure enterprise-grade security, NABH readiness, and scalability

3. SCOPE OF WORK

3.1 Website + Patient Portal (Enterprise)

- Advanced responsive hospital website
- Patient self-service login portal
- Online appointment booking and live tracking
- Doctor profiles with real-time availability
- Health blog, news, and patient education section
- Online bill payment and report access

Cost: Rs. 1,00,000 - Rs. 1,50,000

3.2 Doctor Appointment & Queue Management

- Real-time multi-doctor appointment booking
- Smart queue management system
- Auto doctor availability update
- Appointment rescheduling and cancellation
- Department-wise scheduling and token system

Cost: Rs. 1,20,000 - Rs. 1,80,000

3.3 Patient Management - Full EMR / EHR

- Complete patient records (OPD + IPD structure)
- Digital prescriptions with doctor e-signature
- Full medical history timeline
- Lab report, imaging and document management
- Patient portal access for records
- Discharge summary generation
- Chronic disease and follow-up management

Cost: Rs. 1,50,000 - Rs. 2,50,000

3.4 Staff Management + Face Recognition + HRMS

- Full HRMS with payroll integration
- AI face recognition attendance system
- Shift and roster management
- Leave, performance, and appraisal tracking
- Doctor-wise OPD and revenue reports

Cost: Rs. 1,50,000 - Rs. 2,50,000

3.5 Medicine Inventory & Pharmacy System

- Multi-store pharmacy inventory management
- Batch, expiry, and barcode tracking
- Supplier portal and purchase management
- Auto stock replenishment alerts
- Pharmacy billing and integration with OPD/IPD

Cost: Rs. 1,00,000 - Rs. 1,80,000

3.6 Billing, Insurance & TPA Management

- OPD, IPD, and pharmacy billing
- GST-compliant and NABH-ready invoicing
- Insurance and TPA claim management
- Co-payment and deductible calculation
- Revenue cycle and financial reports

Cost: Rs. 1,00,000 - Rs. 1,60,000

3.7 Admin Dashboard (Enterprise Analytics)

- Multi-branch real-time performance dashboard
- Department-wise patient and revenue reports
- Doctor productivity and OPD analytics
- Bed occupancy and IPD management
- Custom report generation and export

Cost: Rs. 80,000 - Rs. 1,50,000

3.8 Mobile Applications (Android / iOS)

- Patient App: Appointment booking, report access, bill payment, notifications, teleconsultation
- Doctor App: Patient records, appointment view, digital prescription, lab reports

Cost: Rs. 2,00,000 - Rs. 4,00,000

4. ADDITIONAL FEATURES

4.1 AI-Based Face Recognition Attendance

- Touchless auto staff check-in/out
- Secure doctor and staff login via face ID

Cost: Rs. 80,000 - Rs. 1,20,000

4.2 AI Analytics & Clinical Insights

- Patient readmission risk alerts
- Disease pattern analysis
- OPD demand forecasting

Cost: Rs. 1,00,000 - Rs. 1,50,000

4.3 Telemedicine (Video Consultation)

- Online doctor-patient video consultation
- Teleconsultation scheduling and payment
- E-prescription post consultation

Cost: Rs. 80,000 - Rs. 1,50,000

4.4 SMS + Email + WhatsApp Integration

- Appointment and discharge alerts
- Lab report delivery via WhatsApp
- Promotional health campaigns

Cost: Rs. 40,000 - Rs. 60,000

4.5 Security, Audit & Compliance System

- Multi-level role-based access control
- Full activity and access audit trail
- AES-256 data encryption
- NABH and HIPAA-ready data handling

Cost: Rs. 50,000 - Rs. 1,00,000

5. TECHNOLOGY STACK (MERN + AI + MOBILE)

- **Frontend:** React.js

- **Backend:** Node.js + Express.js
- **Database:** MongoDB
- **Mobile Apps:** React Native / Flutter
- **AI Integration:** AWS Rekognition / ML models
- **Video:** WebRTC / Twilio (Telemedicine)
- **Hosting:** AWS (EC2, S3, CloudFront, RDS optional)

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 9,00,000 - Rs. 15,00,000+ (One-time)
Infrastructure	AWS Hosting: Rs. 10,000 - Rs. 50,000/month
Infrastructure	Domain + SSL: Rs. 2,000/year
Infrastructure	CDN & Backup: Rs. 20,000/year
Maintenance Cost	Rs. 50,000 - Rs. 1,00,000/month/month

7. PROJECT TIMELINE

Phase	Duration
Planning	2 Weeks
Development	10-14 Weeks
Testing	2-3 Weeks
Deployment	1-2 Weeks
Total Duration	3-4 Months

8. LIMITATIONS

- High initial investment
- Requires proper staff training
- Infrastructure cost is higher
- Third-party API costs are recurring

9. FUTURE UPGRADES

- Integration with government health portals (ABDM/ABHA)
- Insurance & TPA full automation
- IoT device integration (smart health monitoring)

- AI chatbot for 24/7 patient support

10. SUPPORT AND WARRANTY

- 2-3 months free support
- Priority bug fixing
- Dedicated support team and account manager

11. TERMS AND CONDITIONS

- 40% advance payment
- 30% mid-phase
- 30% after completion
- Additional modules charged separately

12. CONCLUSION

This Premium HMS solution provides a **complete digital transformation platform** for hospitals — combining full EMR/EHR, AI-powered features, telemedicine, mobile apps, enterprise analytics, and NABH-ready compliance, ensuring long-term growth, patient satisfaction, and operational excellence.

13. CONTACT DETAILS

Company Name:	Kartsho Digital
Parent Company:	A Venture of Kartsho Enterprises
Phone:	+91 9453134901 / +91 9528660578
Email:	kartshoenterprises@gmail.com
Website:	digital.kartsho.com
Address:	Godhoopatti, Patti, Pratapgarh, Uttar Pradesh - 230134